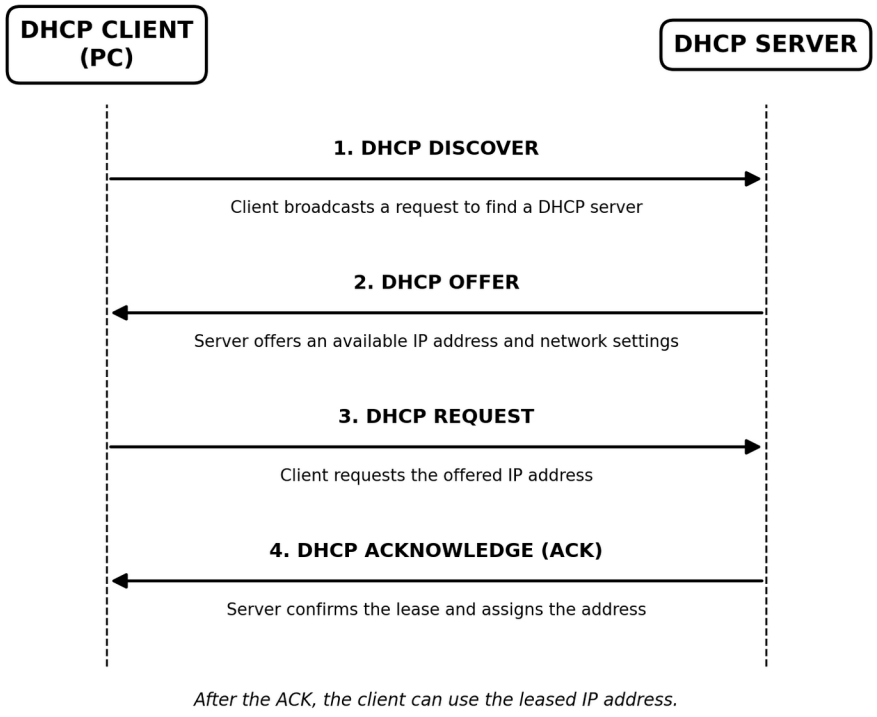
DYNAMIC HOST CONFIGURATION PROTOCOL (DHCP) ASSIGNMENT

1. In Cisco Packet Tracer, set up a network with one DHCP server and two client PCs. Configure automatic IP assignment and verify that each PC receives a different address from the DHCP pool.

A DHCP server automatically assigns IP configuration to devices on a network. In this setup, the DHCP server would be connected to the same LAN as two client PCs. A DHCP pool would be configured with a network range, subnet mask, default gateway, and DNS server. Both PCs would be set to obtain an IP address automatically (DHCP).

Result: When each PC requests network settings, the DHCP server leases an available address from the pool. For example, PC0 may receive 192.168.1.10 and PC1 may receive 192.168.1.11, with the same subnet mask and gateway. The two clients receive different IP addresses so that they can communicate on the network without an address conflict.

2. Draw a sequence diagram showing the DHCP process: Discover, Offer, Request, and Acknowledge, with the sender and receiver labelled.



Explanation: These four messages are known as the DORA process. The client first discovers a DHCP server, the server offers an available address, the client requests that address, and the server acknowledges the lease. After receiving the acknowledgement, the client can use the assigned IP configuration.

3. Change the DHCP pool so that only one IP address is available. Add two PCs and explain what happens when both try to obtain an address.

If the DHCP pool contains only one available IP address, the first PC to request an address can receive that address from the server. When the second PC requests an address, the server has no free address left to offer, so the second PC cannot obtain a valid lease from that pool.

Why only one PC succeeds: A DHCP server cannot lease the same IP address to two clients at the same time because this would cause an IP address conflict. Therefore, only the first client receives the single available address; the other client may show an automatic/private address or remain without a usable IPv4 configuration until an address becomes available.

4. Simulate a Zomato-style restaurant Wi-Fi scenario in Cisco Packet Tracer. Configure DHCP for a laptop and a smartphone (represented by a generic PC), show their assigned IPs, and explain why DHCP is useful.

Scenario: A restaurant provides Wi-Fi for staff devices. A DHCP server is configured with a pool for the restaurant's wireless network. The laptop and smartphone are set to obtain their IP addresses automatically.

Device	Example assigned IP address	Subnet mask	Default gateway
Staff laptop	192.168.10.101	255.255.255.0	192.168.10.1
Smartphone (generic PC)	192.168.10.102	255.255.255.0	192.168.10.1

Why DHCP is useful: DHCP automatically gives each device a unique IP address and the required network settings. This makes it easier for staff devices to join the restaurant network without manually configuring each device, saves time, and reduces configuration errors and IP address conflicts.